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CryoNuSeg: A dataset for nuclei instance segmentation of cryosectioned H&E-stained histological images

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ABSTRACT

Nuclei instance segmentation plays an important role in the analysis of hematoxylin and eosin (H&E)-stained images. While supervised deep learning (DL)-based approaches represent the state-of-the-art in automatic nuclei instance segmentation, annotated datasets are required to train these models. There are two main types of tissue processing protocols resulting in formalin-fixed paraffin-embedded samples (FFPE) and frozen tissue samples (FS), respectively. Although FFPE-derived H&E stained tissue sections are the most widely used samples, H&E staining of frozen sections derived from FS samples is a relevant method in intra-operative surgical sessions as it can be performed more rapidly. Due to differences in the preparation of these two types of samples, the derived images and in particular the nuclei appearance may be different in the acquired whole slide images. Analysis of FS-derived H&E stained images can be more challenging as rapid preparation, staining, and scanning of FS sections may lead to deterioration in image quality.

In this paper, we introduce CryoNuSeg, the first fully annotated FS-derived cryosectioned and H&E-stained nuclei instance segmentation dataset. The dataset contains images from 10 human organs that were not exploited in other publicly available datasets, and is provided with three manual mark-ups to allow measuring intra-observer and inter-observer variabilities. Moreover, we investigate the effects of tissue fixation/embedding protocol (i.e., FS or FFPE) on the automatic nuclei instance segmentation performance and provide a baseline segmentation benchmark for the dataset that can be used in future research.

A step-by-step guide to generate the dataset as well as the full dataset and other detailed information are made available to fellow researchers at https://github.com/masih4/CryoNuSeg.

1. Introduction

Digital pathology enables the acquisition, management and sharing of information retrieved from stained digitised tissue sections from patient-derived biopsies in a digital environment. This offers many benefits including image interpretation by remotely located specialists or further use of the samples for scientific purposes [2]. An additional advantage of digitised samples is their utilisation for computer-mediated quantitative image analysis [10]. In combination, digital pathology and computational image analysis are expected to significantly improve clinical practice, for instance through second opinion systems supporting the work of pathologists [3].

Examination of hematoxylin and eosin (H&E)-stained tissue sections can reveal important information about individual cells and their functional status [5]. Consequently, judgement of these histopathological images remains the "gold standard" in diagnosing a variety of diseases including almost all types of cancer. Nuclei morphology, shape, type, count, and density are the key components in the evaluation process of H&E-stained tissue images. To extract these features automatically with a computer-based method, nuclei instance segmentation is required [17].

Nuclei instance segmentation masks should provide exact boundaries around each individual nucleus and also distinguish overlapping or touching objects. While many computer-based nuclei instance

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segmentation methods have been proposed in the literature, supervised deep learning (DL)-based approaches are the most promising [16,17,31]. However, they require fully annotated datasets to be able to train the neural network model, while annotated data is also required to quantitatively evaluate any model's performance. While manual labelling of nuclei by biomedical experts is considered the gold standard method to create nuclei instance segmentation ground truths, this is very time demanding and may suffer from intra-observer and inter-observer variabilities.

A number of fully manually annotated nuclei instance segmentation datasets of H&E-stained images have been formerly introduced as shown in Table 1. In addition, PanNuke [7] is a recent semi-automatically created dataset of formalin-fixed and paraffin-embedded samples with nearly 200,000 nuclei from 19 different organs where the nuclei masks were first created automatically and then revised by humans.

While these datasets contain H&E-stained images and corresponding instance segmentation ground truths, important information regarding the utilised tissue processing approach is missing. There are two main types of tissue processing methods that can be applied before H&E-staining of sections, namely formalin (chemical)-fixation and paraffin-embedding (FFPE) of tissue samples, and preparation of frozen samples (FS). The general workflows to obtain H&E-stained sections using these two methods are shown in Fig. 1.

H&E staining of FFPE samples is the more widely used approach. Here, samples are fixed with formalin and embedded in blocks of paraffin wax. Thin slices of these blocks are then stained and used for diagnostic purposes. While samples derived form this workflow are very durable, the entire procedure is time-consuming, taking hours to days [23]. On the other hand, preparation of sections from FS is very fast, requiring between 5 and 20 min [23]. Here, samples are rapidly frozen (for structure preservation) and then a cryo-microtome is used to section them.

Following H&E-staining, samples are investigated under a microscope. FS staining is usually required in oncological surgeries, where rapid and online diagnosis and decisions are important for further processes. Distinguishing benign from a malignant lesions, online cancer grading, determining the borders around abnormal tissues, and accessing the invasion status of lesions are some applications where FS are used [12]. A 92–98% diagnostic accuracy with images derived from FS can be achieved by experienced pathologists. However, in some challenging cases, different diagnostic outcomes are reported for FFPE- and FS-derived H&E stained images. This is mainly due to technical problems of rapid sample preparation in FS-derived samples which may lead to poor quality of the derived whole slide images (WSIs). Due to the same technical issue, the nuclei appearance may be different in FS-derived samples compared to FFPE-derived samples. More condensed nuclear chromatin or nuclear ice crystals are some of the artefacts of FS-derived stained sections [12]. These artefacts cannot only affect the interpretation by medical experts but may also have a significant impact on the performance of DL-based algorithms for nuclei instance segmentation. In general, due to insufficient numbers of available FS-derived H&E-stained WSIs, only few studies have analysed FS-derived H&E-stained tissue sections [1,15,26,27] and, to our knowledge, no research has been conducted to explicitly analyse nuclei instance segmentation of FS-derived H&E-stained images.

The main contributions of our work in this paper are as follows:

- We present CryoNuSeg, the first fully annotated nuclei instance segmentation dataset based solely on FS-derived H&E-stained images. Additionally, our dataset comprises images from human organs that have not been used in formerly released datasets.
- The nuclei in our dataset are labelled manually by two experts. Moreover, one of the annotators re-labelled the entire dataset. Therefore, we have three sets of manual mark-ups for the dataset, allowing for the measurement of inter-observer and intra-observer variabilities.
- The entire dataset is made publicly available to fellow researchers.
- We investigate the effect of tissue fixation/embedding protocol (i.e., FS or FFPE) on instance segmentation performance using a state-of-the-art DL-based nuclei instance segmentation algorithm.
- We create a baseline segmentation benchmark for the dataset, which can be used in future studies.

Our dataset is publicly available on the Kaggle platform.[1] We provide step-by-step descriptions of sample selection, sample preparation, and generation of segmentation masks in this paper and in the publicly available repository at https://github.com/masih4/CryoNuSeg.

2. Method

2.1. Dataset

The Cancer Genome Atlas (TCGA)[2] contains more than 30,000 WSIs from more than 50 human organs and tissues. Using the filtering options provided in TCGA, we selected FS-derived H&E-stained images acquired at 40x magnification and chose images from organs not present in other publicly available datasets. With the help of a senior cell biologist from the Medical University of Vienna, we selected 30 WSIs from 10 different human organs (three WSIs per organ), namely the adrenal gland, larynx, lymph node, mediastinum, pancreas, pleura, skin, testis, thymus, and thyroid gland. To maximise data variability, we also aimed to include samples from different scanning centres, different disease types and different sexes. Using QuPath,[3] we extracted image patches of a fixed size of 512×512 pixels (one image patch per WSI) while aiming to extract the patches from the most representative parts of the WSIs. All image patches were extracted as RGB 8-bit images in the (uncompressed) TIFF format. To perform manual segmentation, we used ImageJ[4] and its pre-built region of interest (ROI) manager tool. Further technical step-by-step descriptions of WSI selection, WSI patch extraction by QuPath, and manual annotation with ImageJ are detailed in the released GitHub repository.

Two annotators, a biologist (Annotator 1) and a bioinformatician (Annotator 2) were employed for manual segmentation of the nuclei. While the biologist has a deeper prior knowledge of cell biology, before commencing the annotation task, both annotators received the same biological background information including information about the variety of appearance of nuclei in different tissues. Both annotators were then given the dataset and a document containing technical instructions for the segmentation. Annotator 1 also performed a second round of manual mark-ups with a gap of three months between the two annotations.

In the entire dataset, Annotator 1 identified and segmented 7596 in the first annotation round (the number reported in Table 1) and 8044 nuclei in the second round, while Annotator 2 segmented 8251 nuclei. Detailed information regarding the number of segmented nuclei in each organ and by each annotator is listed in Table 2.

Given the higher degree of experience in the field of cell biology, the manual nuclei segmentations of Annotator 1 are used in this paper for evaluation in our experiments. More specifically, all experiments on the CryoNuSeg dataset are performed using the first round annotations of Annotator 1 as a basis.

Besides creating labelled and binary masks, we also created additional auxiliary segmentation masks that can be useful in training supervised DL-based approaches. These auxiliary masks include binary segmentation masks generated by removing touching borders, distance

[1] https://www.kaggle.com/ipateam/segmentation-of-nuclei-in-cryosectioned-he-images.

[2] https://portal.gdc.cancer.gov/repository.

[3] https://qupath.github.io/.

[4] https://imagej.nih.gov/ij/.

Table 1
Publicly available datasets for nuclei instance segmentation. Note that the Kaggle Data Science Dataset contains both H&E-stained bright-field and fluorescence microscopic images while all others contain only H&E-stained images. MoNuSeg = Multi-Organ Nuclei Segmentation; MoNuSAC = Multi-Organ Nuclei Segmentation and Classification; CoNSep = Colorectal Nuclear Segmentation and Phenotypes; CPM: Computational Precision Medicine; TNBC: Triple Negative Breast Cancer; CRCHisto: Colorectaladeno Carcinomas. TCGA: The Cancer Genome Atlas; UHCW: University Hospitals Coventry and Warwickshire. The last row of the table shows the statistics of the proposed CryoNuSeg dataset (the # nuclei there refers to the number of segmented nuclei by Annotator 1 in their first round of manual mark-ups).

Dataset	# image tiles	# nuclei	magnification	# organs	tile size [pixels]	source
Kumar et al. [17]	30	21,623	40 ×	7	1000 × 1000	TCGA
MoNuSeg [16]	44	28,846	40 ×	9	1000 × 1000	TCGA
MoNuSAC [28]	209	31,411	40 ×	4	81 × 113 to 1422 × 2162	TCGA
CoNSeP [9]	41	24,319	40 ×	1	1000 × 1000	UHCW
CPM-15 [29]	15	2905	40 × , 20 ×	2	400 × 400, 600 × 1000	TCGA
CPM-17 [29]	32	7570	40 × , 20 ×	4	500 × 500 to 600 × 600	TCGA
TNBC [22]	50	4022	40 ×	1	512 × 512	Curie Inst.
CRCHisto [25]	100	29,756	20 ×	1	500 × 500	UHCW
Kaggle Data Science [4]	670	29,464	n/a	n/a	256 × 256 to 1040 × 1388	n/a
Janowczyk [13]	143	12,000	40 ×	1	2000 × 2000	n/a
Crowdsource [11]	64	2532	40 ×	1	400 × 400	TCGA
CryoNuSeg (this study)	30	7596	40 ×	10	512 × 512	TCGA

Table 2
Number of manually segmented nuclei for each organ in the CryoNuSeg dataset and each set of annotations (two annotations by Annotator 1 and one by Annotator 2).

organ	# images	# nuclei		
		Ann 1 (round 1)	Ann 1 (round 2)	Ann 2
adrenal gland	3	338	344	392
larynx	3	622	641	765
lymph node	3	1204	1308	1353
mediastinum	3	1274	1349	1354
pancreas	3	470	548	623
pleura	3	497	515	638
skin	3	457	436	507
testis	3	752	793	687
thymus	3	1521	1646	1483
thyroid gland	3	461	464	449
total	30	7596	8044	8251

maps, and weighted maps that give more weight to the pixels between close nuclei. Such masks have been shown useful for nuclei instance segmentation in former studies [18,22,24,29], and the related codes to create both conventional and auxiliary segmentation masks are made available in our GitHub repository.

Fig. 2 gives some examples of raw image patches together with the resulting conventional and auxiliary segmentation masks. In Fig. 3, we show some inconsistent cases between the segmentation masks of Annotator 1 and Annotator 2 (inter-observer variability) and some mismatched cases between the first and second segmentations of Annotator 1 (intra-observer variability).

To investigate the effects of the tissue fixation/embedding protocol on DL-based nuclei segmentation, we use the MoNuSeg dataset [16] described in Table 1. Information regarding the tissue processing type is not provided in former datasets, however, since TCGA is also the data source of the MoNuSeg dataset, by tracking down the image codes it is possible to retrieve the tissue fixation/embedding protocol (FS or FFPE) used in the MoNuSeg images. Of the 44 image patches in the MoNuSeg dataset, 27 images are from FS-derived sections and 17 from FFPE-derived sections. Based on this, we separate the images to form two subsets, namely MoNuSeg-FS and MoNuSeg-FFPE. We use the same number of images per organ to be able to compare the nuclei segmentation results. For instance, of the nine kidney images in MoNuSeg, seven are FS and two FFPE samples. Thus, to have balanced datasets, we randomly choose two FS kidney images for the MoNuSeg-FS subset. We exclude colon and lung images since their samples are all obtained with the same tissue fixation/embedding protocol. The number of annotated nuclei in the derived MoNuSeg-FS and MoNuSeg-FFPE datasets are 7639 and 7,503, respectively, and are split amongst the organs as listed in Table 3. Some sample images from MoNuSeg-FS and MoNuSeg-FFPE are shown in Fig. 4. It should be noted that the apparent differences in images from the same organ, besides the differences in tissue fixation/embedding protocol, can also be related to differences in staining protocols, scanning systems, and disease types.

2.2. Nuclei segmentation method

We use our recently published state-of-the-art DL-based instance segmentation algorithm [18] as the baseline segmentation model. The general workflow of this method is illustrated in Fig. 5.

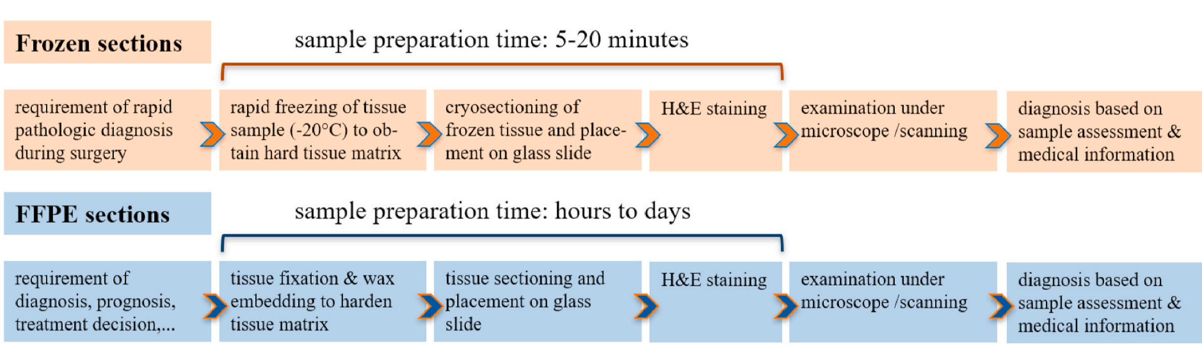

Fig. 1. Preparation workflow of frozen sections (top) and formalin-fixed paraffin-embedded (FFPE) sections (bottom). The fixation and embedding procedure of FFPE samples is much more time consuming compared to the freezing step required to prepare frozen tissue samples.

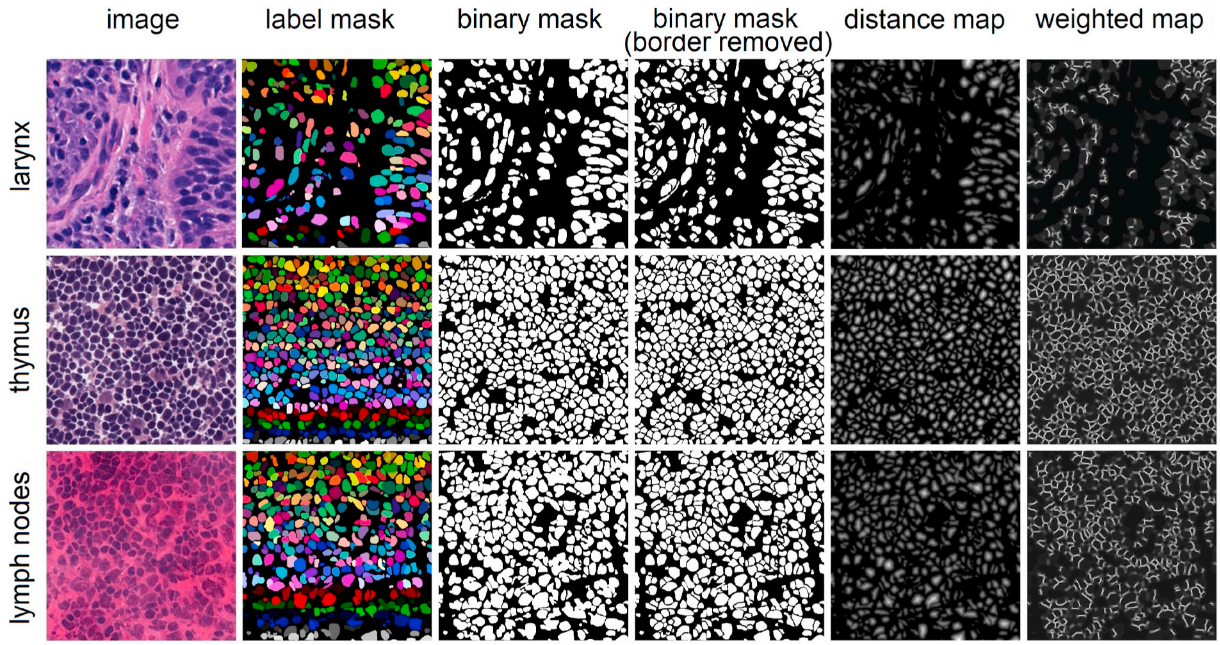

Fig. 2. Sample CryoNuSeg images from three human organs and their corresponding segmentation masks (from first segmentations of Annotator 1). For each sample, we show (from left to right) the raw image patch, the manually labelled nuclei, the binary segmentation mask, the binary mask with touching borders removed, the distance map, and the weighted map.

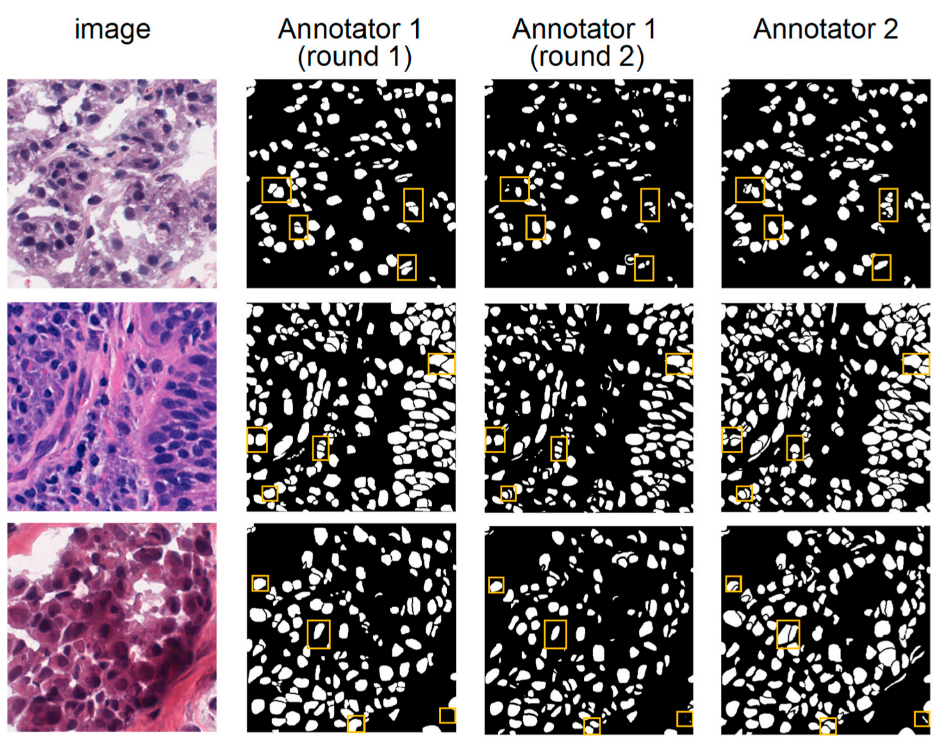

Fig. 3. Examples of deviations in the annotations. Some mismatching annotations exemplifying inter-observer variability (comparison between second/third column and fourth column) and intra-observer variability (comparison between second and third column) are shown in yellow boxes. While there are more mismatched pairs in the masks, we limit the number of mismatched pairs to four for better visualisation.

Images are normalised by mapping the intensity values to the $[0;1]$ range before being fed to the model. Unlike in our original approach in Ref. [18], we do not use any colour normalisation step as pre-processing, while to deal with the colour variations in different images we manipulate the colour schemes of the images by various random augmentation techniques.

The segmentation model consists of two sub-models, which are trained independently. A U-Net [24]-based model (segmentation U-Net) is used to separate foreground and background, while a regression encoder-decoder-based model (distance U-Net) is employed to predict the distance maps for all nuclei instances. Both sub-models contain an encoder with five blocks of convolutional layers, dropout layers and

Table 3
Number of images and nuclei per organ in the MoNuSeg-FS and MoNuSeg-FFPE datasets.

organ	MoNuSeg-FS		MoNuSeg-FFPE	
	# images	# nuclei	# images	# nuclei
breast	3	1267	3	1117
bladder	1	677	1	342
prostate	4	1716	4	1475
brain	1	465	1	249
stomach	1	1165	1	1390
liver	3	1461	3	1282
kidney	2	888	2	1648
total	15	7639	15	7503

max-pooling layers and a decoder with five blocks of convolutions layers, dropout layers and transposed convolutional layers. Similar to the original U-Net implementation, both sub-models use skip connections, which connect different blocks of the encoder path to the decoder path. The main differences between the two sub-models are the last activation layer and the utilised loss functions. For the segmentation U-Net, a sigmoid activation is used in the last layer while for the distance U-Net, a linear activation is applied. We use a combination of Dice loss and binary cross-entropy to train the segmentation U-Net, whereas a mean squared error loss function is used to train the distance U-Net. After training the two sub-models, their results are merged to form the instance segmentation masks.

To obtain the final instance segmentation masks, we first apply a Gaussian smoothing filter on the predicted distance maps to prevent false local maxima detection, with the kernel size of the filter determined from the segmentation U-Net results based on the average predicted nuclei size. Then, the local maxima are derived from the smoothed distance maps and are used as seed points for a watershed algorithm. We use the results from the segmentation U-Net as the mask for the watershed method to determine all background pixels.

We utilise the full-size (512×512) images in both training and testing. Each of the sub-models is trained for 30 epochs with a learning rate scheduler (the initial learning rate of 0.01 is halved after every 8 epochs). We set the batch size to 8 and apply various augmentation techniques in the training phase including random horizontal and vertical flipping (with a probability of 0.5), random 90, 180, and 270° rotations (with a probability of 0.5), random brightness and contrast shift (brightness and contract shift limit of 0.15 with a probability of 0.4), and random adaptive histogram equalisation (tile grid size of 8 with the probability of 0.5) [19,20]. In the inference phase, we apply two post-processing steps on the final segmentation results, which remove very small instances from the segmentation masks (objects with areas containing less than 20 pixels) and fill holes inside detected objects (using morphological operations).

2.3. Evaluation

To evaluate the segmentation performance, we employ three evaluation indexes [9,14,17], namely the Dice score, aggregate Jaccard index (AJI), and panoptic quality (PQ) score. The Dice score evaluates the general performance of semantic segmentation, while both AJI and PQ measure instance segmentation performance. Compared to AJI, PQ is a more robust measure and does not suffer from over-penalisation. To detect statistically significant differences in the obtained results, we use a two-sided Wilcoxon signed-rank test [8].

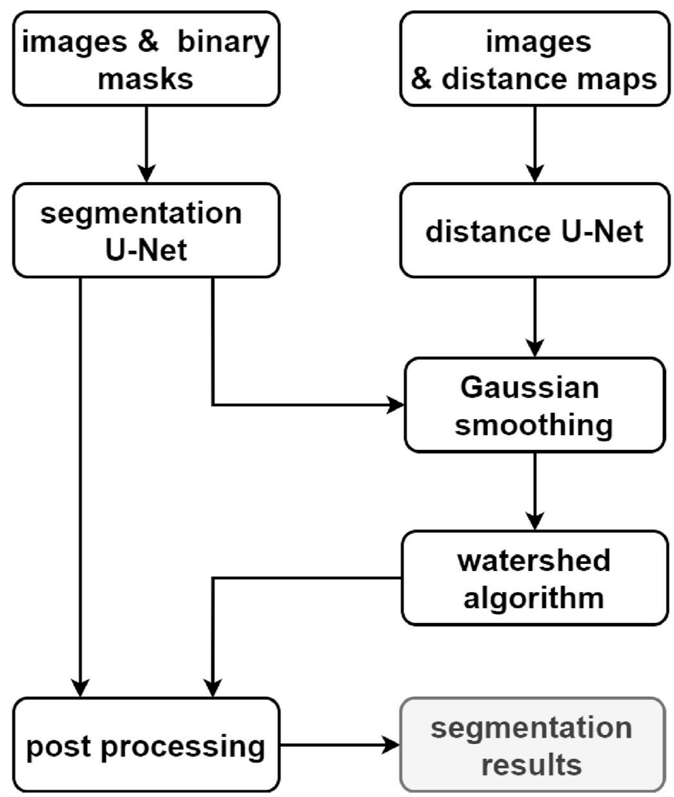

Fig. 5. Overview of the employed instance segmentation algorithm [18].

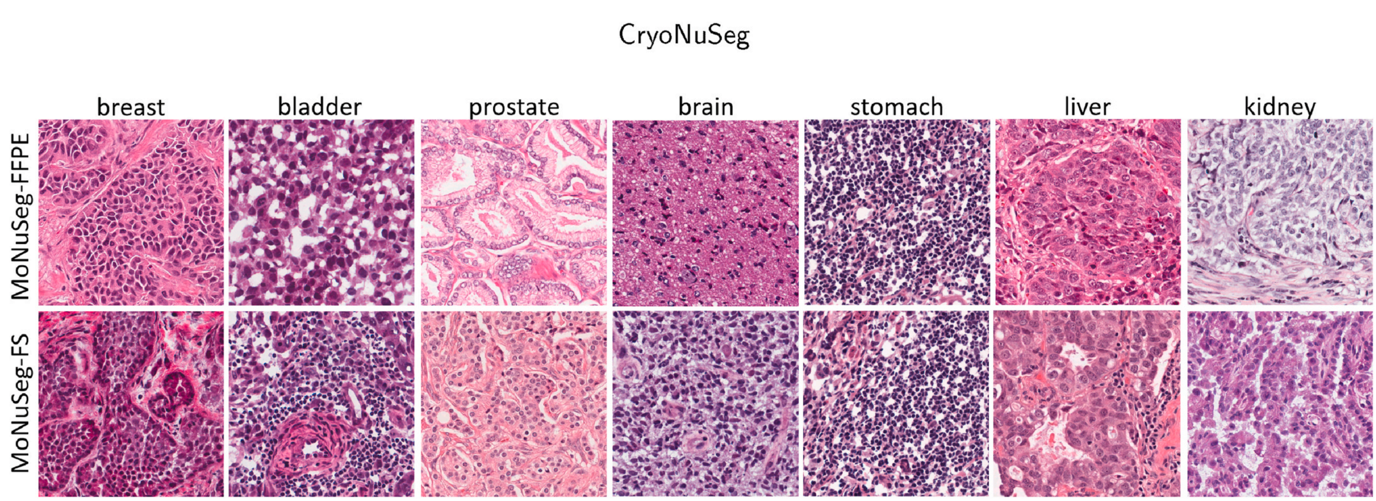

Fig. 4. Sample images from the derived MoNuSeg-FFPE and MoNuSeg-FS datasets.

2.4. Implementation

The code to generate the CryoNuSeg segmentation masks from the ImageJ ROI files is implemented in Matlab 2018a, while for the segmentation model we use the Keras DL framework (version 2.3.1). All experiments are performed on a single workstation with an Intel Core i7-8700 3.20 GHz CPU, 32 GB of RAM, and a TITIAN V NVIDIA GPU card with 12 GB of installed memory.

3. Results

We first use the MoNuSeg-FS dataset as well as the MoNuSeg-FFPE dataset as training sets and employ the entire CryoNuSeg dataset as the test set. We perform an identical training scheme for training both models as described in Section 2.2. The main aim of this experiment is to investigate the effect of tissue fixation/embedding protocol on the performance of the nuclei segmentation model. The obtained results are given in Table 4.

We perform an additional experiment using the combined MoNuSeg-FS and MoNuSeg-FFPE datasets as training data and the entire CryoNuSeg dataset as test set. This results in an average Dice score of 79.1$\pm$ 5.2%, average AJI of 51.8 $\pm$ 6.4%, and average PQ score of 48.2$\pm$ 7.3%. A statistical test comparing the results derived from the combined MoNuSeg-FS/MoNuSeg-FFPE dataset gives p-values of 0.0196, 0.0001, and 0.0001 for Dice score, AJI, and PQ score, respectively, when compared to the results from MoNuSeg-FS, and p-values of 0.0068 (Dice), 0.0005 (AJI), and 0.0001 (PQ) when compared to the results from the MoNuSeg-FFPE dataset.

In the next experiment, we use the same segmentation model but perform 10-fold cross-validation (10CV) on the CryoNuSeg dataset to obtain segmentation results. In each fold, 27 images from nine organs are used for training, and the trained model is then evaluated on the hold-out test organ. The motivation behind this experiment is to create a baseline segmentation benchmark that should prove useful for future studies based on the introduced CryoNuSeg dataset. The results of this experiment are shown in Table 5.

To confirm superior performance of the employed two-stage U-Net algorithm in comparison to either a single segmentation U-Net model or a single distance U-Net model, we perform an ablation study and report the obtained 10CV results in Table 6.

We perform an additional 10CV experiment where in each fold 27 images from nine organs of the CryoNuSeg dataset together with the combined MoNuSeg-FS/MoNuSeg-FFPE dataset are used for training and the images of the hold-out organ from the CryoNuSeg dataset are used as the test set (i.e., there are 57 training images and 3 test images in each fold). The results for this are an average Dice score of 80.2$\pm$ 4.8%, an average AJI of 52.9 $\pm$ 5.7%, and an average PQ score of 49.4$\pm$ 6.5%.

Table 5
10CV segmentation results on CryoNuSeg dataset in terms of Dice score, AJI, and PQ score (based on the segmentation masks of Annotator 1 in the first round of mark-ups).

test organ	Dice (%)	AJI (%)	PQ (%)
adrenal gland	78.17	53.49	48.30
larynx	81.65	59.70	54.50
lymph node	81.56	53.54	50.79
mediastinum	84.87	54.10	50.73
pancreas	74.31	44.84	37.75
pleura	75.49	46.49	40.02
skin	74.75	47.84	40.78
testis	84.27	50.49	47.51
thymus	85.90	56.46	52.83
thyroid gland	81.81	58.20	53.48
average	80.3 $\pm$ 4.3	52.5 $\pm$ 5.0	47.7 $\pm$ 6.1

While in the former experiments, we choose the segmentation masks from Annotator 1 (first round of manual mark-ups) as the ground truth, in our next experiment, we investigate the inter-observer variability between the two annotators. For this, we compare the manual segmentation masks generated by Annotator 2 to those from Annotator 1 (first round) with the latter being used as ground truth. The results are reported in Table 7.

Last not least, to measure intra-observer variability, we compare the first- and second-round of segmentation masks of Annotator 1 and show the results in Table 8.

4. Discussion

In this paper, we introduce the first fully manually annotated nuclei segmentation dataset based on FS-derived H&E-stained sections. Further, we investigate the impact of the tissue fixation/embedding procedure on the segmentation performance of a state-of-the-art DL-based nuclei segmentation algorithm. Our dataset can be used alongside other publicly available datasets to train supervised machine learning-based approaches or can be used as a stand-alone benchmark to

Table 6
Comparison, in terms of 10CV segmentation results on CryoNuSeg, of the employed two-stage U-Net algorithm with single-stage segmentation U-Net and single-stage distance U-Net.

method	Dice (%)	AJI (%)	PQ (%)
segm. U-Net	79.4	38.8	36.7
segm. U-Net + watershed	79.3	47.8	40.4
dist. U-Net + watershed	74.7	48.6	37.5
two-stage U-Net	80.3	52.5	47.7

Table 4
Segmentation results on CroNuSeg when trained on MoNuSeg-FS and MoNuSeg-FFPE, respectively. The results, in terms of Dice score, AJI, and PQ score, are based on the manual segmentation masks from Annotator 1 (first round of mark-ups) as ground truth. The reported p-values in the bottom row show the results of the statistical significance test when comparing the two training sets.

test organ	Dice score (%)		AJI (%)		PQ score (%)	
	MoNuSeg-FS	MoNuSeg-FFPE	MoNuSeg-FS	MoNuSeg-FFPE	MoNuSeg-FS	MoNuSeg-FFPE
adrenal gland	78.31	80.80	51.64	57.11	47.01	52.08
larynx	80.54	80.80	57.61	57.68	54.56	54.65
lymph node	79.71	80.00	49.62	49.90	49.07	49.18
mediastinum	83.24	83.34	50.56	50.84	47.19	47.85
pancreas	67.43	66.06	38.04	37.18	32.43	31.91
pleura	71.74	69.20	42.96	41.74	37.31	36.22
skin	72.96	72.33	45.82	45.73	40.29	40.40
testis	81.08	82.20	48.54	50.21	44.20	44.96
thymus	84.50	84.66	53.36	53.76	48.35	48.97
thyroid gland	80.11	80.46	55.80	56.83	48.19	51.62
average	78.0 $\pm$ 5.5	78.0 $\pm$ 6.4	49.4 $\pm$ 5.9	50.1 $\pm$ 6.8	44.9 $\pm$ 6.5	45.8 $\pm$ 7.4
***p*-value**	0.6800		0.1529		0.0719	

Table 7
Segmentation results, in terms of Dice score, AJI, and PQ score, from comparing the manual segmentation masks from Annotators 2 to those from Annotator 1 (first round) to show inter-observer variability.

test organ	Dice (%)	AJI (%)	PQ (%)
adrenal gland	78.08	53.16	49.77
larynx	83.68	62.53	58.39
lymph node	82.38	57.80	54.80
mediastinum	84.09	58.34	54.36
pancreas	67.14	40.66	35.93
pleura	69.74	45.09	42.62
skin	74.06	47.89	46.55
testis	83.86	55.37	51.12
thymus	85.88	62.00	58.49
thyroid gland	80.52	58.36	56.57
average	78.9 ± 6.5	54.1 ± 7.3	50.9 ± 7.3

Table 8
Segmentation results, in terms of Dice score, AJI, and PQ score, from comparing Annotator 1's manual segmentation masks of the first round to those of the second round to show intra-observer variability.

test organ	Dice (%)	AJI (%)	PQ (%)
adrenal gland	84.83	62.25	55.44
larynx	84.92	67.25	63.63
lymph node	84.77	63.22	59.46
mediastinum	84.67	62.40	57.32
pancreas	78.27	54.63	50.71
pleura	82.17	61.00	53.37
skin	78.95	55.16	53.69
testis	86.21	61.42	54.12
thymus	87.84	67.77	60.90
thyroid gland	85.69	68.79	61.66
average	83.8±3.1	62.4± 4.8	57.0±4.2

evaluate the performance of nuclei segmentation methods.

The description of the dataset, the steps to generate it, and the related implementation codes to create conventional and auxiliary segmentation masks from the ImageJ ROI files are available in the published Github repository, while the dataset itself is available on the Kaggle website. With Kaggle's cloud-based computational resources, it is possible to use the dataset and develop instance segmentation models directly on the Kaggle website. We have implemented an example kernel in the Jupyter notebook-based environment on the Kaggle platform based on the well-known U-Net algorithm with and without watershed post-processing [24,30]. This kernel should prove helpful to show how to use the dataset on the Kaggle platform and is publicly available on the dataset page under the notebook section.[5]

The results in Table 4 show the impact of the tissue fixation/embedding procedure on instance segmentation performance based on training models derived from the MoNuSeg-FS and MoNuSeg-FFPE datasets, which contain only FS-derived and only FFPE-derived H&E-stained images, respectively. While FS- and FPPE-derived images can never be obtained from the exact same sample, we have tried to minimise other data variability differences by using the same number of images per organ and samples of the same organs (breast, bladder, prostate, brain stomach, liver, and kidney) which results also in roughly the same number of segmented nuclei (7639 for MoNuSeg-FS and 7503 for MoNuSeg-FFPE). As shown in Table 4, the overall and organ-wise segmentation results are very competitive (except for AJI and PQ score for adrenal gland) in both cases (i.e., when trained on MoNuSeg-FS and MoNuSeg-FFPE, respectively). The Wilcoxon signed-rank test yields p-values larger than 0.05 for all three evaluation measures (Dice score, AJI, and PQ score). Thus, we find nuclei instance segmentation performance to not be significantly affected by the employed tissue fixation/embedding procedure. As there are rather large segmentation performance differences based on AJI and PQ score for the adrenal gland for the MoNuSeg-FS and MoNuSeg-FFPE datasets, further research with a larger training sample of adrenal gland images is required to investigate the impact of tissue fixation protocol on the nuclei instance segmentation for this type of tissue.

Training on the combined MoNuSeg-FS/MoNuSeg-FFPE experiment leads to a slight but statistically significant improvement in segmentation performance, especially in terms of PQ score, which results from the larger training set (30 images compared to 15) in this experiment. Further research is required to investigate the impact of the tissue fixation/embedding protocol on other histological image analysis tasks such as gland/tumour segmentation or WSI classification/grading.

As a segmentation benchmark for our CryoNuSeg dataset, we perform 10-fold cross validation whereby the images of nine organs are used for training while testing on the tenth and repeating the process so that each organ is once used for testing. This separation method had two advantages over a random division. First, the segmentation results reported in Table 5 better describe the generalisation ability of the model, since in each fold an unseen test organ is used for the evaluation. Second, for future studies that use this dataset for developing instance segmentation models, identical folds can be easily created to fairly compare segmentation results. The results from the experiment, where MoNuSeg-FS and MoNuSeg-FFPE are additionally used for training, show, as expected, a slight improvement in the segmentation performance.

As the results of the ablation study in Table 6 show, merging the results from the two sub-models of the employed segmentation algorithm, leads to improved instance segmentation performance as observed by higher AJI and PQ scores) while a small improvement in terms of Dice score is also noticeable. These results confirm the rationale of selecting the algorithm as our baseline for a benchmark on the dataset.

Comparing the results from Tables 4 and 5, we can observe that segmentation performance improves when training on CryoNuSeg images compared to training on MoNuSeg images (either MoNuSeg-FS or MoNuSeg-FFPE). The difference is statistically significant as confirmed by a pair-wise statistical test for each of the utilised evaluation indices, with the obtained p-values are well below 0.05 for all six cases (three evaluation measures each for MoNuSeg-FS vs. CryoNuSeg and MoNuSeg-FFPE vs. CryoNuSeg). This could be related to the larger number of training images in the CryoNuSeg dataset compared to the MoNuSeg-FS/MoNuSeg-FFPE images, although all three datasets have roughly the same number of annotated nuclei. Another reason could be related to the inter-observer variability since for CryoNuSeg, a single person (Annotator 1 in our experiments) annotated both training and test images, while for the MoNuSeg-FS and MoNuSeg-FFPE images different annotators were used for training and test images.

The CryoNuSeg dataset is provided with segmentations from two annotators (a biologist and a bioinformatician), with both instructed identically on how the segmentations should be obtained. To measure the inter-observer variability, Table 7 compares the segmentation masks of the two annotators. While in an ideal scenario a perfect match would be achieved, the obtained results show that there is a significant difference. The difference is relatively small in terms of Dice score but more evident for AJI and PQ score. This indicates that the overall agreement between the two annotators to distinguish background and foreground is much better than on distinguishing touching or overlapping nuclei. From visual inspection of the manual segmentation masks from the two annotators, it can be noticed that Annotator 2 has a tendency to over-segment as shown by some examples in Fig. 3 and as demonstrated by the higher number of identified nuclei compared to Annotator 1. The issue of inter-observer variability was also observed in a subset of the MoNuSeg dataset where the agreement between two annotators (based on AJI) was reported to be 65% [16].

Since two segmentations are available from Annotator 1, we can

[5] https://www.kaggle.com/ipateam/u-net-with-binary-labels.

measure intra-observer variability as reported in Table 8. While a perfect match would represent the ideal situation, the results in Table 8 show that there is a significant difference for all three evaluation indexes (and much more evident for AJI and PQ score). By comparing the average and organ-wise results in Tables 7 and 8, it can be observed that inter-observer variability has a larger impact on the segmentation performance in comparison to intra-observer variability.

Fuzzy or unclear borders between touching nuclei, folded tissues and other acquisition artefacts, manual annotation errors in the nuclei borders, and sensitivity loss of the annotators due to fatigue are some parameters that can cause inter-observer and intra-observer variability issues. These problems can be partially resolved by removing vague areas in the manual segmentation masks [28], but this requires extra supervision and time. The masks obtained from individual experts can be used to obtain new types of ground truth using e.g. through majority rules or to investigate how an annotator's expertise impacts the training or evaluation of an individual automatic nuclei instance segmentation approach [6,21].

We can also compare the manual segmentation results of the second annotator from Table 7 with the 10CV results from Table 5 to benchmark the automated segmentation algorithm against a human annotator. From this, we can conclude that the results are relatively close, with the statistical tests yielding *p*-values of 0.055 for Dice score, 0.052 for AJI, and 0.003 for PQ score, which suggests that the baseline segmentation method we provide is within reach of the performance of a manual annotator (although the PQ score results are statistically significantly different).

5. Conclusions

In this paper, we have introduced the first fully manually annotated FS-derived H&E-stained nuclei segmentation dataset. Our CryoNuSeg dataset is publicly available to fellow researchers together with extensive documentation and sample implementations. It can thus be used for developing and comparing nuclei instance segmentation algorithms. We have also provided a baseline segmentation benchmark founded on a state-of-the-art DL-based approach for this purpose. In addition, we have investigated the impact of the tissue fixation/embedding method on the segmentation performance, while further work is planned to investigate and compare the impact of tissue fixation and embedding techniques on different histological image analysis tasks. We hope that the dataset will prove useful for the community and are looking forward to its use in future studies.

Declaration of competing interest

There are no conflicts of interest to disclose for publication of this paper.

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